Illustration 6.3: A missile is launched from the army ship to attack and another from the land to intercept it. The loci of the missiles are shown in figure.

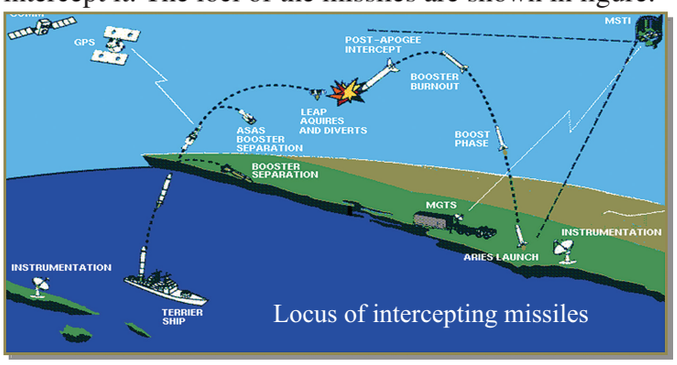
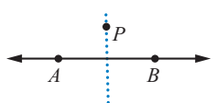
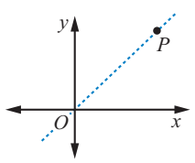
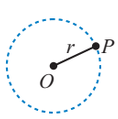


Figure 6.5

Locus of missiles play a vital role in many wars. During the Gulf War (2 Aug1990-28 Feb 1991), Iraq attacked Israeli cities with Scud missiles. To defend from Scud attack, Israel used Patriot missiles to shoot down enemy missiles. To launch a satellite or space shuttle successfully, the determination of path plays an crucial role in space research.

An equation in the two variables x and y will ordinarily be satisfied by infinitely many pair of real value of x and y. Every such pair is called a real solution of the equation. Each real solution of the equation will have its graph. The collection of all these graphs is called the locus of the given equation.

The following table shows some important loci in mathematics

A moving point ***P*** under the given condition	Graph	Name of the path
A point P moves such that it is equidistant from two fixed points A and B	 Figure 6.6	Perpendicular bisector of the line segment ***AB***
A point P moves such that it is equidistant from two fixed lines ox and oy	 Figure 6.7	Angle bisector of the angle$\angle xoy$
A point P moves equidistant from a fixed point O	 Figure 6.8	Circle

Now let us discuss the ways of finding the locus of the points. The equation of the locus is the relation that exists between the coordinates of all the points strictly lying on the path.

Procedure for finding the equation of the locus of a point

(i) If we are finding the equation of the locus of a point P, assign coordinates, say (h, k) to P

(ii) Express the given conditions as equations in terms of the known quantities and unknown parameters.

(iii) Eliminate the parameters, so that the resulting equation contains only h, k and known quantities.